

Result

Size

Time

Cycles

GPU

SM Frequency

Process

Attributes

Current

556 - knn_kernel

(8192, 1, 1)x(256, 1, 1)

4.83 s

2,82,28,09,560

0 - Tesla T4

585.00 Mhz

[52016] bench_knn

Summary

Details

Source

Context

Comments

Raw

Session

Compare

Tools

View

Export

GPU Speed Of Light Throughput

GPU Throughput Breakdown

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor.

Compute (SM) Throughput [%]	52.41	Duration [s]	4.83
Memory Throughput [%]	49.93	Elapsed Cycles [cycle]	2,82,28,09,560
L1/TEX Cache Throughput [%]	99.86	SM Active Cycles [cycle]	2,81,70,32,531
L2 Cache Throughput [%]	0.04	SM Frequency [Mhz]	585.00
DRAM Throughput [%]	0.00	DRAM Frequency [Ghz]	4.99

Latency Issue

This workload exhibits low compute throughput and memory bandwidth utilization relative to the peak performance of this device. Achieved compute throughput and/or memory bandwidth below 60.0% of peak typically indicate latency issues. Look at [Scheduler Statistics](#) and [Warp State Statistics](#) for potential reasons.

Key Performance Indicators

Compute Throughput Breakdown

Memory Throughput Breakdown

SM: Mio Pq Read Cycles Active [%]	52.41	L1: Data Pipe Lsu Wavefronts [%]	49.93
SM: Mio Pq Write Cycles Active [%]	7.80	L1: Lsu Writeback Active [%]	23.94
SM: Issue Active [%]	5.51	L1: Lsuin Requests [%]	3.30
SM: Inst Executed [%]	4.16	L1: Data Bank Writes [%]	0.38
SM: Inst Executed Pipe Lsu [%]	3.30	L1: Data Bank Reads [%]	0.37
SM: Inst Executed Pipe Cbu Pred On Any [%]	3.16	L1: M Xbar2l1tex Read Sectors [%]	0.05
SM: Pipe Alu Cycles Active [%]	2.23	L1: M L1tex2xbar Req Cycles Active [%]	0.05
SM: Pipe Fma Cycles Active [%]	1.52	L2: T Sectors [%]	0.04
SM: Mio Inst Issued [%]	1.10	L2: Lts2xbar Cycles Active [%]	0.04
SM: Mio2rf Writeback Active [%]	0.78	GPU: Compute Memory Access Throughput Internal Activity [%]	0.04
SM: Inst Executed Pipe Uniform [%]	0.68	L2: Xbar2lts Cycles Active [%]	0.04
SM: Inst Executed Pipe Adu [%]	0.00	L2: T Tag Requests [%]	0.04
SM: Pipe Tensor Cycles Active [%]	0	L2: D Sectors [%]	0.01
SM: Inst Executed Pipe Fp16 [%]	0	DRAM: Cycles Active [%]	0.00
SM: Pipe Shared Cycles Active [%]	0	DRAM: Dram Sectors [%]	0.00
SM: Pipe Fp64 Cycles Active [%]	0	L1: Texin Sm2tex Req Cycles Active [%]	0.00
IDC: Request Cycles Active [%]	0	L2: D Sectors Fill Device [%]	0.00
SM: Memory Throughput Internal Activity [%]	0	L2: D Sectors Fill Systemem [%]	0.00
SM: Instruction Throughput Internal Activity [%]	0	L1: F Wavefronts [%]	0.00
SM: Inst Executed Pipe Xu [%]	0	L1: Data Pipe Tex Wavefronts [%]	0
SM: Inst Executed Pipe Tex [%]	0	L1: Tex Writeback Active [%]	0
SM: Inst Executed Pipe Ipa [%]	0	L2: D Atomic Input Cycles Active [%]	0
		GPU: Compute Memory Request Throughput Internal Activity [%]	0

Memory Workload Analysis

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between those units (Max Bandwidth), or by reaching the maximum throughput of issuing memory instructions (Mem Pipes Busy).

Memory Throughput [Kbyte/s]	760.95	Mem Busy [%]	49.93
L1/TEX Hit Rate [%]	58.42	Max Bandwidth [%]	23.94
L2 Hit Rate [%]	99.87	Mem Pipes Busy [%]	52.41
L2 Compression Input Sectors [sector]	n/a	L2 Compression Success Rate [sector]	n/a

Occupancy

% Occupancy Graphs

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	100	Block Limit Registers [block]	10
Theoretical Active Warps per SM [warp]	32	Block Limit Shared Mem [block]	42
Achieved Occupancy [%]	99.81	Block Limit Warps [block]	4
Achieved Active Warps Per SM [warp]	31.94	Block Limit SM [block]	16

Impact of Varying Register Count Per Thread

Impact of Varying Block Size

Impact of Varying Shared Memory Usage Per Block

To customize your report even further, you might want to learn about [custom sections](#) and [writing your own rules](#). You might also want to consider [adding individual metrics](#).

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